

ML Project Charter

Predictive Exit Interview System

Project name	Predictive Exit Interview System
Prepared for	HR Analytics / People Operations
Dataset	IBM HR Analytics Employee Attrition & Performance (Kaggle)
Charter version	v1.0
Status	Draft — for stakeholder review

1. Problem Statement

Voluntary employee attrition is typically identified only after an employee has already resigned, at which point retention interventions are no longer possible. This project develops an automated framework to identify high-risk attrition candidates before they resign, allowing HR and people managers to intervene proactively — through engagement, compensation review, role changes, or workload adjustments — and improve overall retention rates.

2. Business Objective

Reduce voluntary attrition by flagging employees at elevated resignation risk early enough for HR business partners to act, replacing today's reactive exit-interview process with a proactive, data-driven early-warning system.

Success metrics (KPIs)

Metric	Target	Rationale
Recall on attrition class (Yes)	≥ 0.60	Missing a true leaver is costlier than a false alarm — recall is prioritized over raw accuracy given the ~16% class imbalance.
Precision on attrition class (Yes)	≥ 0.35	Keeps HR intervention workload manageable; too many false positives erodes trust in the tool.
ROC-AUC	≥ 0.75	Overall discriminative ability across risk thresholds.
Retention lift (business KPI, post-deployment)	5-10% reduction within 2 quarters	The ultimate business outcome the model is meant to drive.

3. Scope

In scope

- Batch model that scores current employees on attrition risk using historical HR attributes (compensation, tenure, satisfaction, role, travel, etc.).
- Feature importance analysis to surface the top drivers of attrition for HR to act on directly (not just prediction, but explanation).
- A defined ML pipeline from raw data ingestion through model serving (see Workflow Pipeline Diagram).

- Model versioning and reproducibility via Git + DVC.

Out of scope (phase 1)

- Real-time streaming scoring (phase 1 is scheduled batch scoring, e.g. monthly).
- Automated intervention actions (the model informs HR; it does not trigger offers, raises, or transfers automatically).
- Free-text exit interview NLP analysis (reserved for a future phase).

4. Dataset

IBM HR Analytics Employee Attrition & Performance dataset (Kaggle, published by IBM). 1,470 employee records, 35 attributes including demographics, compensation, satisfaction scores, tenure, and the binary target Attrition (Yes/No). Baseline class distribution is approximately 84% No / 16% Yes, which informs the class-imbalance handling strategy (stratified sampling, class weighting) noted in section 6.

Note: this is a synthetic, IBM-published sample dataset used to validate the modeling approach. Production deployment requires retraining on the organization's own HRIS data under appropriate data governance and employee privacy review.

5. Stakeholders & Roles

Role	Responsibility
HR Business Partners	Act on risk flags; provide domain validation of feature importance
People Analytics / Data Science	Build, validate, and maintain the model and pipeline
HRIS / IT	Data access, integration, and secure serving infrastructure
Legal / Compliance	Review for fairness, bias, and employee privacy compliance
Executive sponsor (CHRO office)	Approves charter, owns retention KPI outcomes

6. Modeling Approach & Baseline Results

A baseline Random Forest classifier was trained on the Kaggle dataset (80/20 stratified train/test split, class-balanced weighting) to validate feasibility ahead of full pipeline build-out. Results below establish the floor the production pipeline must beat.

Metric	Value
Accuracy	0.83
Precision (Attrition = Yes)	0.40
Recall (Attrition = Yes)	0.09
ROC-AUC	0.79

Interpretation: Recall on the minority (Yes) class is currently well below the 0.60 target in section 2, which is expected from an untuned baseline on an imbalanced dataset. Priority next steps: threshold tuning, SMOTE/oversampling or cost-sensitive learning, and hyperparameter search — see Roadmap in section 8.

7. Top Attrition Risk Drivers (Feature Importance)

The full feature importance table is delivered as a separate artifact. The top drivers identified by the baseline model are summarized here for charter readers:

Rank	Feature	Importance
1	MonthlyIncome	0.078
2	Age	0.069
3	DailyRate	0.057
4	TotalWorkingYears	0.057
5	YearsAtCompany	0.049
6	MonthlyRate	0.048
7	DistanceFromHome	0.047
8	HourlyRate	0.045
9	OverTime	0.045
10	NumCompaniesWorked	0.041

8. Risks, Assumptions & Ethical Considerations

- **Bias/fairness risk:** Age, Gender, MaritalStatus, and DistanceFromHome are present in the dataset and correlate with protected characteristics in some jurisdictions. Model must be audited for disparate impact before any HR action is taken on its output, and Legal/Compliance sign-off is required before production use.
- **Label leakage risk:** Features like YearsSinceLastPromotion or StockOptionLevel may reflect management decisions already influenced by informal attrition-risk signals — this should be reviewed during feature engineering.

- **Self-fulfilling prophecy risk:** Flagging an employee as "high risk" and treating them differently could itself influence their decision to leave. Risk scores must remain confidential and be used only for supportive interventions, never punitive ones.
- **Data assumption:** The Kaggle dataset is synthetic/sample data; production model must be retrained on the organization's own historical data before deployment.
- **Consent/privacy:** Employees should be informed that HR analytics uses aggregated workforce data for retention planning, per applicable data protection policy.

9. High-Level Roadmap

Phase	Milestone	Target duration
1. Discovery	Charter sign-off, data access, governance review	2 weeks
2. Pipeline build	Ingestion, preprocessing, feature engineering, versioning (see pipeline diagram)	3 weeks
3. Modeling	Baseline → tuned model, meets KPI targets in section 2	3 weeks
4. Validation	Bias/fairness audit, stakeholder review, pilot on 1 department	3 weeks
5. Deployment	Batch scoring in production, HR dashboard rollout	2 weeks
6. Monitor & iterate	Track retention lift KPI, retrain cadence, drift monitoring	Ongoing

10. Deliverables

- ML Project Charter (this document)
- Feature Importance Table (standalone artifact)
- Proposed Workflow Pipeline Diagram (standalone artifact)
- Versioned model pipeline code (Git repository)
- Model card documenting performance, limitations, and fairness audit results (post-validation phase)

Prepared as part of an HR analytics mini-project exercise using the IBM HR Analytics Attrition dataset. Figures in sections 6–7 are derived from an actual baseline model trained on this dataset for illustrative purposes.